Journal of Nuclear Materials 458 (2015) 394–405

Contents lists available at ScienceDirect

Journal of Nuclear Materials

journal homepage: www.elsevier.com/locate/jnucmat

Characterising the U–Nd–O miscibility gap by an experimental and a theoretical approach

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HIGHLIGHTS

- Existence of a miscibility gap in $(U_{1-y},Nd_y)O_2$ oxide at room temperature was found.
- A thermodynamic model on the U–Nd–O system was developed using the CALPHAD method.
- The influence of the oxygen potential of the system during fabrication was analysed.

ARTICLE INFO

Article history:
Received 21 May 2014
Accepted 15 December 2014
Available online 23 December 2014

ABSTRACT

Recent studies performed on $(U_{1-y},Nd_y)O_2$ highlighted the existence of a miscibility gap in this mixed oxide. The objective of this work is to describe this miscibility gap by means of two types of approaches: experimental and theoretical. The first approach involves a set of SEM analyses combined to XRD characterisations. The second approach consists in the thermodynamic assessment of the U–Nd–O ternary system using the CALPHAD method. A good agreement is obtained between the calculated and experimental data for Nd/(U + Nd) ratios from 3% to 28%. Both results show that this miscibility gap appears preferentially when the O/M ratio is less than two and when the Nd atomic fraction of the sample is higher than 0.04. This confirms that the existence of the miscibility gap in neodymium UO_2 doped ceramics is consistent with the phase diagram calculated on the basis of a thermodynamic model. Finally, we discuss the conditions leading to the appearance of the miscibility gap with respect to the fabrication process.

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1. Introduction

A great number of fission products (FP) and minor actinides are created in nuclear oxide fuels due to the nuclear fissions produced by neutron irradiation, with many of them considered to be soluble in the crystallographic structure of the ceramic (Nd, La, Gd, Pu, Np, etc.) [1]. It is important to investigate the effects of incorporating these FPs into the fluorite fuel lattice since the presence of additional species in the fuel matrix could affect the overall characteristics of the fuel, and consequently, its behaviour. Among these soluble species, neodymium deserves special attention, because it is the most abundant soluble fission product. As a consequence, it can be used as a "model" fission product in order to study the effect of incorporating fission products into the UO_2 fuel matrix.

Despite the traditional assumptions affirming that Nd has a high solubility level in UO_2 [2], as well as some recent studies like that of Venkata [3], a miscibility gap was recently detected in the U–Nd–O ternary system. This miscibility gap has been identified through various experimental studies. Some of them revealed the existence of two face-centred cubic (FCC) phases by XRD characterisation [4,5]. The other study revealed an additional peak in the results of Raman spectrometry [4,6]. Nevertheless, the most irrefutable proof of this miscibility gap was obtained by HT-XRD analyses: the $(U_{1-y},Nd_y)O_2$ sample was single-phase at high temperature, but a temperature-reversible phase transition occurred during cooling with the appearance of two fluorite phases [7].

The appearance of two FCC phases is not a new phenomenon in lanthanide-doped oxide fuels. As an example, Lorenzelli [8] also detected a phase transition in temperature for the U–Ce–O system by means of HT-XRD characterisation. Fujino worked on the U–Eu–O system and detected a two-phase domain in $(U_{1-y},Eu_y)O_2$ from $0.09 < x(Eu) < 0.15$ [9]. Ohmichi also studied many $(U_{1-y},L_y)O_2$

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http://dx.doi.org/10.1016/j.jnucmat.2014.12.061
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systems (L = lanthanide) and detected two FCC phases in some cases [5]. It seems that the appearance of a miscibility gap in these systems is a complex phenomenon. Currently, the conditions leading to its appearance in lanthanide-doped oxide fuels have yet to be identified. Moreover, such a miscibility gap seems to be very sensitive to environmental parameters: it may have a narrow area in the phase diagram.

Therefore, the aim of this work is to characterise and analyse this miscibility gap in the U–Nd–O in order to obtain an accurate description of its extent in composition and temperature. Two complementary approaches were chosen to perform this research: an experimental procedure and a theoretical approach. The experimental method consists in performing scanning electron microscopy (SEM) analyses coupled with X-ray diffraction (XRD) characterisation; the latter is followed by a structural determination based on a Rietveld refinement. The theoretical approach consists in thermodynamically assessing the U–Nd–O ternary system by using a model developed with the compound energy formalism (CEF) [10–12] supported by the Calphad method.

2. Method

2.1. Experimental characterisation

2.1.1. Samples

Samples used in this work were $(U_{1-y},Nd_y)O_2$ non-irradiated ceramics prepared in the LBF hot laboratory at the CEA/Cadarache centre. The samples with different "y" values were fabricated by mixing UO_2 and Nd_2O_3 powders, before pressing them at 450 MPa and finishing with sintering under He with 5% H_2 at 1923 K for 72 h. The XRD analysis showed no signs of hexagonal Nd_2O_3 at the end of the procedure [6]. The contents of Nd and U in each sample are reported in Table 1. The D sample is a special case since it was subjected to an additional thermal treatment under reducing conditions, as detailed in [7].

2.1.2. Experimental procedure

At a first step, a fragment of each sample was taken and the cross section was polished. These fragments were observed by scanning electron microscopy (SEM). An SEM XL 30 – ESEM6 FEG equipment was used for energy-dispersive X-ray spectroscopy examinations (EDX).

The SEM images were taken with a back-scattered electron (BSE) detector with a magnification of 8000× at 15 kV. In addition, a qualitative elementary analysis by EDX was performed under the same conditions for neodymium and uranium.

In parallel, the samples were characterised by XRD. These analyses were performed using a Bragg Brentano θ–θ Bruker D8 Advance X-ray diffractometer with a Lynx Eye detector, the X-ray being provided by a conventional generator (Cu, $K_{\alpha 1} + K_{\alpha 2}$ radiation). The acquisition was performed with a step of 0.04° and a counting time of 1s/step within the range of 20–120° (2θ).

2.1.3. Structural characterisation

The Rietveld method was employed to analyse the XRD data and to completely characterise the detected phases. This method enabled us to obtain the lattice parameter, the phase proportions and more specially the neodymium and uranium content in each phase [13].

A Split Pearson PVII profile function was chosen to fit the experimental points. The background was determined by a 6-order Chebyshev polynomial taking into consideration a $1/x$-type background. The 2-theta and zero-displacement experimental errors were corrected.

Each detected phase was modelled with a fluorite (FCC) structure. Concerning the atomic positions, U^{4+} and Nd^{3+} cations were introduced in the (0,0,0) site and O^{2-} anions in the (¼,¼,¼) site. Based on this model, the site-occupancy factor and the isotropic thermal parameter were then refined. The position coordinates were not refined.

The refinement was performed with the TOPAS software [14]. The R-weighted pattern factor "R_{wp}" was used as the criterion of fit.

Table 1
Chemical formula of $(U_{1-y}Nd_y)O_2$ samples. The exact stoichiometry in oxygen of the samples was not determined.

Samples	
A	$(U_{0.97}Nd_{0.03})O_2$
B	$(U_{0.91}Nd_{0.09})O_2$
C	$(U_{0.87}Nd_{0.13})O_2$
D	$(U_{0.72}Nd_{0.28})O_2$

2.2. Thermodynamic modelling

The U–Nd–O system was thermodynamically assessed by the Calphad method [15]. With the aim of obtaining an accurate representation of this system and to be able to analyse the miscibility gap from a thermodynamic point of view, the whole ternary phase diagram is modelled. However, efforts focussed on calculating the area delimited by the UO_2, UO_3 and Nd_2O_3 oxides because this is where the miscibility gap is located according to our experiments. The computer software support for the calculations was Thermocalc Version S [16].

The ternary phase diagram was calculated by two steps. In the first step, the binary phase diagrams uranium-neodymium and neodymium-oxygen were modelled, with optimisation of the Gibbs energy of the compounds and solutions. The second step thus involved the U–Nd–O ternary phase diagram calculation from the calculated binaries Nd–O, U–Nd in the present work and from the binary U–O system, which has been previously modelled by Guéneau [12]. This step includes also the optimisation of ternary interaction parameters in the fluorite phase on the basis of experimental data available on the $(U_{1-y}Nd_y)O_2$ phase using the PARROT module [15].

This type of three sublattice model with ionic species was initially developed by Guéneau [12] to calculate different binary, ternary or even quaternary oxide fuels. In this work, we have extended the model to describe the U–Nd–O system by incorporating and modelling all the new phases related to this system, e.g. those containing Nd. All the Gibbs energy functions are detailed in Appendix A.

A preliminary version of the thermodynamic model to calculate the U–Nd, Nd–O and U–Nd–O phase diagrams has already been described in [17] .It will not be detailed in this article. Nevertheless the models and assessed parameters are reported in Appendices A and B respectively.

3. Results

3.1. Experimental results

3.1.1. SEM and EDX analysis

The SEM macrographs and corresponding EDX Nd scans are shown in Fig. 1a and b for the samples A and C respectively. The SEM images show a characteristic UO_2-type ceramic that is well homogenised. The pore sizes in the samples seem to increase with the Nd content; however, the number of pores seems to decrease.

Whereas the SEM images exhibit a homogenised fuel, this is not the case for the EDX acquisitions in the cases of samples B, C and D.

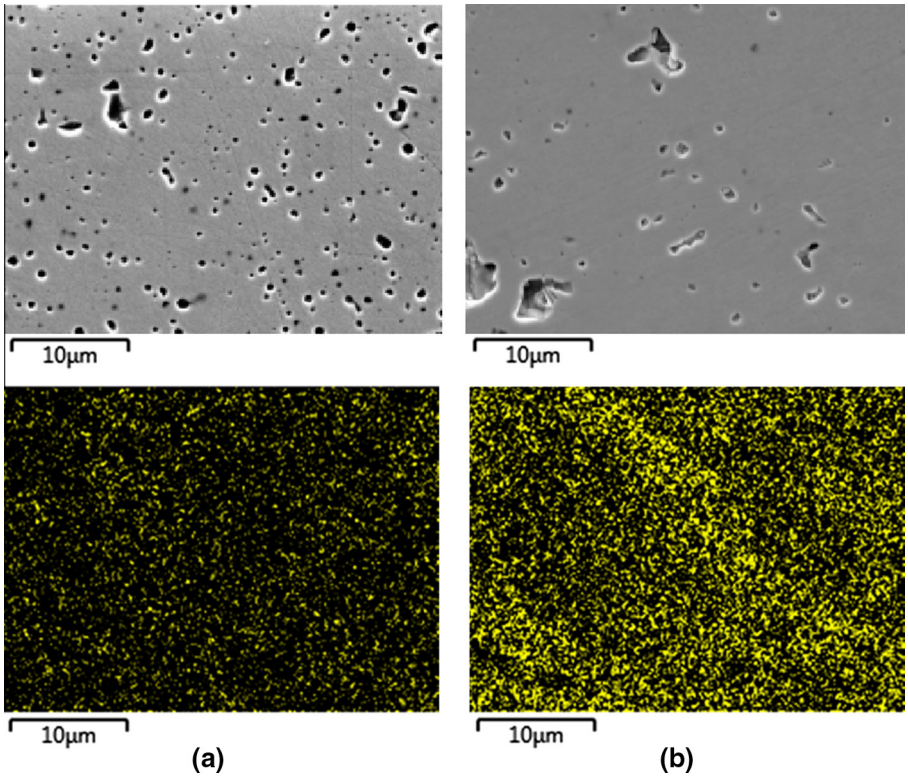

Fig. 1. SEM images and EDX neodymium scans. (a) $U_{0.97}Nd_{0.03}O_2$ (sample A), (b) $U_{0.87}Nd_{0.13}O_2$ (sample C).

Some areas are more enriched in Nd than other areas for these ceramics as is the case for sample C in Fig. 1b.

3.1.2. XRD analysis

The XRD patterns obtained for samples A and C in 74–77° range are shown in Fig. 2a and b, including the peaks (3,3,1) of the UO_2 FCC crystallographic structure.

In Fig. 2(a), it can be observed that for sample A (smallest Nd content), the most homogenous regarding EDX scans, a single peak is observed (containing Kα1 and Kα2 diffraction lines), corresponding to a FCC solid solution with a lattice parameter of 5.470 Å. This is consistent with the typical value considered for non-irradiated and non-doped UO_2 [2]. However, the results obtained for samples B, C and D are different. These samples exhibit two FCC peaks (FCC 1 + FCC 2) instead of a single one (as illustrated in Fig. 2b for sample C).

At the lowest Nd content (sample B), the second FCC peak (FCC 2) appears as a small diffraction peak on the left side of the main peak (FCC 1). As the Nd content increases, this second peak grows and becomes more evident. This behaviour is reflected in the lattice parameters: the mismatch between the two lattice parameters in each sample also increases with the Nd content, as shown in Fig. 3.

The results of the Rietveld refinement are shown in Table 2 for all the studied compositions. The Nd content of the "FCC 2" phase is higher than in the "FCC 1" phase.

3.1.3. Estimation of the O/M ratio and tielines

The oxygen content of each phase was estimated using the Nd values determined from the Rietveld refinement (Table 2). This procedure differs from the one we presented in [7], which uses a model based on ionic radius.

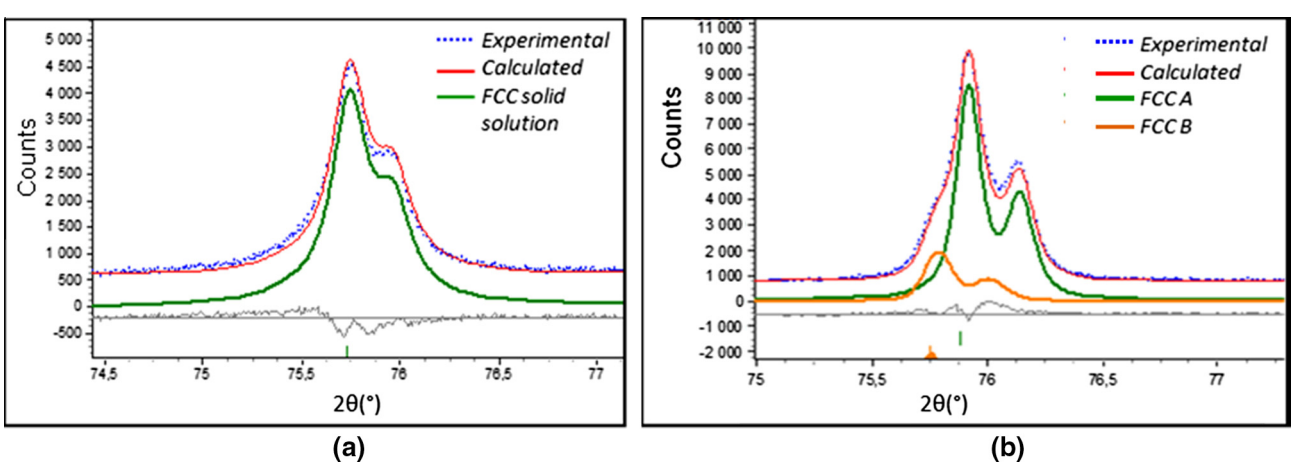

Fig. 2. XRD diffraction patterns: (a) $U_{0.97}Nd_{0.03}O_2$ (Sample A), (b) $U_{0.87}Nd_{0.13}O_2$ (Sample C).

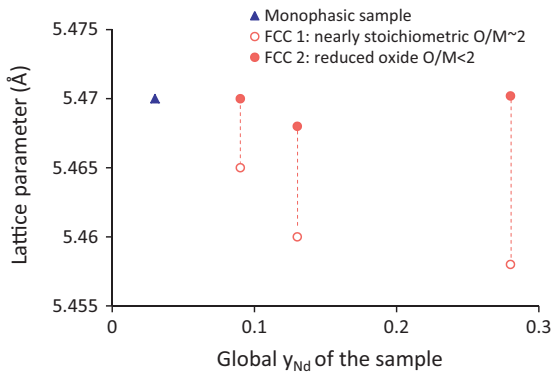

Fig. 3. Lattice parameter variation versus neodymium content in the FCC phases for samples A, B, C and D.

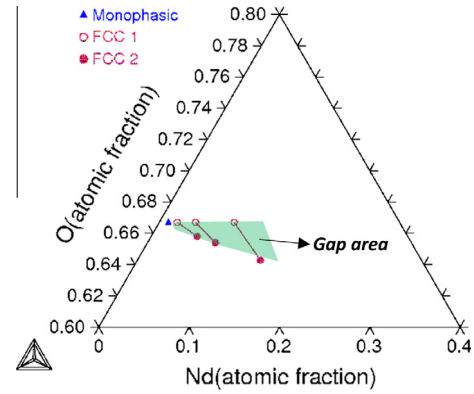

Fig. 4. Representation of experimental results in a ternary phase diagram.

The procedure used in this work basically consists of three steps. In (a), a possible stoichiometry in oxygen was proposed for each phase based on the analysis of the lattice parameter variations. In (b), a chemical form was established for each phase. In the last step (c), the chemical equation was solved using the Nd contents and the O/M was obtained. Finally, we checked if the global Nd content on the sample calculated from these results is consistent was those listed in Table 1. The procedure is detailed below.

3.1.3.1. Oxygen stoichiometry of each FCC phase. Fig. 3 shows the lattice parameter variation for both FCC phases as a function of the overall Nd content in the samples.

For FCC 1, the lattice parameter decreases with the Nd content. For FCC 2, the lattice parameter does not seem to be influenced by the Nd content and remain almost constant. Moreover, FCC 2 values for the lattice parameters are higher than the FCC 1 values in all the samples.

In order to determine which phase has the highest oxygen content, the variation in lattice parameter can be compared with analogue systems such as "U–Ce–O" [8] and "U–Pu–O" [18] in the miscibility gap area: in these systems, the phase "1" shows a decrease in the lattice parameters with the additive content and the other, and the phase "2" shows higher lattice parameters than phase "1", which remain almost constant or increase with additive content. In these systems, the phase "1" has been identified to be a nearly stoichiometric one (slightly hypo-stoichiometric in oxygen), while the phase "2" is a reduced one with a low oxygen to metal ratio (it has been demonstrated that the reduction of UO_2 into UO_{2-x} leads to an increase in the lattice parameter [5]). Thus, if we consider that the miscibility gap in U–Nd–O presents the same behaviour as these systems, it can be established that in our samples FCC 1 is nearly stoichiometric ($O/M \approx 2$) and FCC 2 is hypo-stoichiometric ($O/M < 2$).

3.1.3.2. Chemical form of the system. Based on the previous reasoning, the O/M is nearly equal to 2 in FCC 1. This means that the incorporation of Nd^{3+} is compensated by the oxidation of U^{4+} to U^{5+}. In FCC 2, the O/M is less than two which means that there is an oxygen deficiency. It is assumed that the incorporation of Nd^{3+} is only compensated by the formation of oxygen vacancies (without any change in the uranium valence) [7]. This assumption is consistent with both our proposed model with three sublattices $(U^{3+},U^{4+},U^{5+},Nd^{3+})(O^{2-},Va)_2(O^{2-},Va)$ for thermodynamic modelling, explained in Appendix A, and the associated results obtained from the thermodynamic model presented below.

Hence, the following chemical representations recommended for each phase are as follows:

$$\text{FCC 1}: \left(U^{4}_{1-2y}+,U^{5}_{y}+,Nd^{3+}_{y}\right)O_{2-x} \quad (\text{with } x \sim 0 \text{ due to stoichiometry}) \tag{A}$$

$$\text{FCC 2}: \left(U^{4+}_{1-y},Nd^{3+}_{y}\right)O_{2-x} \quad (\text{with } x = y/2 \text{ due to electroneutrality conditions}) \tag{B}$$

Table 2
Lattice parameter and neodymium content in the FCC phases of samples A, B, C and D using the Rietveld method.

Sample	A $(U_{0.97}Nd_{0.03})O_2$	B $(U_{0.91}Nd_{0.09})O_2$		C $(U_{0.87}Nd_{0.13})O_2$		D $(U_{0.72}Nd_{0.28})O_2$	
Structure	FCC	FCC 1	FCC 2	FCC 1	FCC 2	FCC 1	FCC 2
Phase proportion (%)	100	69	31	89	11	80	20
Lattice parameters (error ≈ 5%)	5.470	5.465	5.470	5.460	5.468	5.458	5.471
y_{Nd} (error ≈ 5%)	0.03	0.06	0.15	0.12	0.22	0.25	0.41
Rwp	5.2	4.7		4.5		9.2	

Table 3
Estimated chemical composition for each FCC phase in samples A, B, C and D.

	Sample A $(U_{0.97}Nd_{0.03})O_2$	Sample B $(U_{0.91}Nd_{0.09})O_2$		Sample C $(U_{0.87}Nd_{0.13})O_2$		Sample D $(U_{0.72}Nd_{0.28})O_2$	
	FCC 1	FCC 1	FCC 2	FCC 1	FCC 2	FCC 1	FCC 2
Chemical form	Eq. (A)	Eq. (A)	Eq. (B)	Eq. (A)	Eq. (B)	Eq. (A)	Eq. (B)
y_{Nd} (error ≈ 5%)	0.03	0.06	0.15	0.12	0.22	0.25	0.41
x	0	0	$0.075(y_{Nd}/2)$	0	$0.11(y_{Nd}/2)$	0	$0.205\ (y_{Nd}/2)$
x(Nd)	0.010	0.020	0.051	0.040	0.073	0.083	0.136
x(U)	0.323	0.313	0.291	0.293	0.270	0.250	0.221
x(O)	0.667	0.667	0.658	0.667	0.654	0.667	0.643

Table 4
Comparison between global Nd contents of the sample from fabrication and global Nd contents calculated from Rietveld results for each phase.

Sample	A $(U_{0.97}Nd_{0.03})O_2$	B $(U_{0.91}Nd_{0.09})O_2$		C $(U_{0.87}Nd_{0.13})O_2$		D $(U_{0.72}Nd_{0.28})O_2$	
Global Nd wt% from fabrication	1.71	5.13		7.80		17.17	
wt% Nd of each phase	0.02	0.03	0.08	0.07	0.13	0.15	0.26
Mass fraction of each phase	1.00	0.69	0.31	0.89	0.11	0.80	0.20
Global Nd wt% from Rietveld	1.62	4.88	7.35	16.86			
Difference (%)	5.45	4.79		5.74		1.81	

Table 5
Crystallographic data for elements U, Nd and compound Nd_2O_3.

Element or compound	Temperature range	Space group
α-U	298–1209 K	Orthorhombic A20 (*Cmcm*) [33]
β-U	1209–1579 K	Tetragonal ($P4_2/mnm$) [33,34]
γ-U	1579–1670 K	Cubic BCC (*Im*-3*m*) [33]
α-Nd	298–1126 K	Hexagonal DHCP($P6_3/mmc$) [33,34]
β-Nd	1126–1294 K	Cubic BCC (*Im*-3*m*) [33]
Nd2O3_A	298–2379 K	Hexagonal (*P*-3*m*1) [22,34]
Nd2O3_H	2379–2477 K	Hexagonal ($P6_3/mmc$) [22]
Nd2O3_X	2379–2577 K	Cubic (*Im*-3*m*) [22]

3.1.3.3. Estimations of phases composition. The chemical compositions of each phase can be obtained based on the expressions of the chemical forms given in Eqs. (A) and (B), as presented in Table 3.

Our estimated compositions are reported in a schematic ternary phase diagram in Fig. 4. The two-phase samples are represented by two points in the phase diagram, connected by the corresponding tieline. The miscibility gap zone is the area delimited by these points, represented by the green region.

3.1.3.4. Verification of the global composition of the sample. When the y_{Nd} and x values obtained from our estimation are converted into mass units, the mass proportion of each phase obtained from Rietveld (Table 2) can be used to obtain the mass balance of the

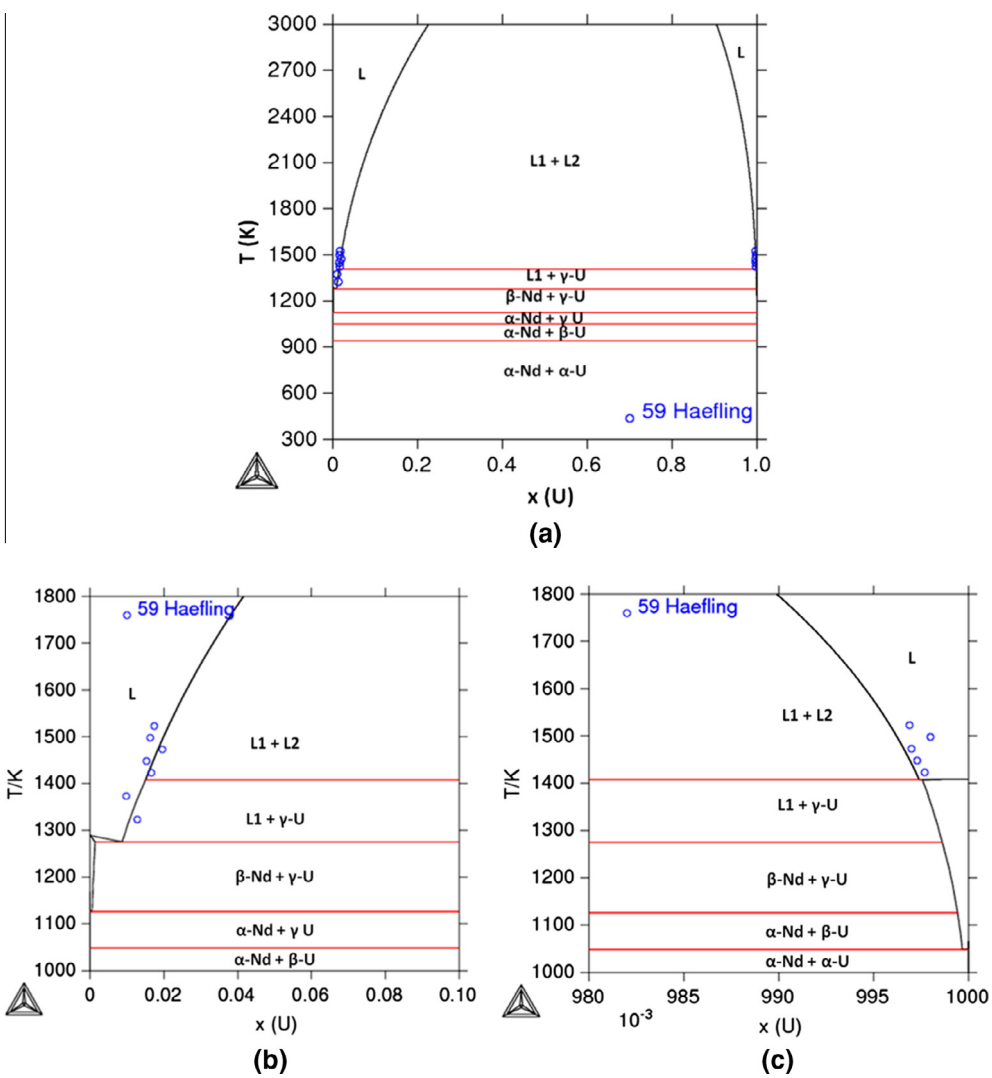

Fig. 5. Calculated U–Nd binary phase diagram: (a) for $0 < x(Nd) < 1$, (b) detail for $0 < x(Nd) < 0.1$, (c) detail for $0.98 < x(Nd) < 1$.

system and to check if the global composition of the sample (obtained from composition of each phase and proportion phases) are in agreement with the global composition of the sample known from fabrication.

The mass balance is given by:

$$\text{Total Nd wt\%} = w_1\text{Nd wt\%}1 + w_2\text{Nd wt\%}2 \quad \text{(C)}$$

Total Nd wt% = Total Nd mass percentage of the sample
w_i = Mass fraction of the "i" phase in the sample (mass i/total mass)
Nd wt%i = Nd mass percentage of the "i" phase

The results are shown in Table 4. There is a difference of about 5% between the total Nd wt% known from fabrication and the total Nd wt% from Eq. (C), which verifies the accuracy of our estimation.

3.1.4. Conclusion

The experimental procedure made it possible to characterise the miscibility gap by means of two different techniques, EDX and XRD. As observed, this phenomenon becomes visible in $(U_{1-y}Nd_y)O_2$ ceramics for x(Nd) contents as low as 0.03. The Rietveld refinement made it possible to estimate the Nd content for each phase within the miscibility gap. Furthermore, the corresponding O/M ratio for each FCC phase was estimated on the basis of the lattice parameter variation and the behaviour observed in similar systems also exhibiting a miscibility gap. Consequently, it can be concluded that in the case of a miscibility gap in FCC uranium-neodymium mixed oxide fuels, one phase is almost stoichiometric and depleted in Nd, while the other is a reduced phase with a higher Nd content.

3.2. Thermodynamic assessment

The uranium-neodymium and neodymium–oxygen binary phase diagrams were assessed in [17]. Crystallographic data on α-U, β-U, γ-U, α-Nd, β-Nd and Nd_2O_3 (A, H, X) phases are displayed in Table 5. The model of U–O system derived by Guéneau [12] was chosen. The models developed for these three binary systems were used in order to calculate the ternary system U–Nd–O. These models are described in Appendix A.

3.2.1. Uranium-neodymium system

The calculated phase diagram of the uranium-neodymium system is shown in Fig. 5a–c. No compounds are reported in literature. No data on the Nd solubility in α-U (orthorhombic), β-U (tetragonal) and γ-U (cubic BCC) or on U solubility in α-Nd (hexagonal) and β-Nd (cubic) exist. The solubility data of γ-U in liquid neodymium and the miscibility gap boundaries in the liquid were taken from [19]. These experimental data were obtained by an annealing treatment under He of an equal volume of U and Nd. As is the case for most lanthanides, the solubility of U in Nd and of Nd in U is very small (e.g. solubility of 0.2 at.% of Nd in U is observed at 1175 °C). As there is very little knowledge on the U–Nd system, the phase diagram is tentative, exhibiting a large miscibility gap in the liquid and BCC (β-Nd, γ-U) phases. Consequently, positive interaction parameters between uranium and neodymium were introduced in the bcc and liquid phases. The model and the assessed parameters are reported in the Appendices A and B.

3.2.2. Neodymium-oxygen system

The Neodymium-oxygen phase diagram calculated is showed in Fig. 6a and b. The experimental oxygen solubility data in neodymium (orthorhombic α-Nd, cubic β-Nd and liquid) were measured by Sano et al. [20], using solid state electrotransport and metal-oxide equilibrium methods. As discussed before, the Nd–O system contains only one intermediate oxide, Nd_2O_3, which is considered as a stoichiometric compound [21]. The Gibbs energy functions for the three forms of Nd_2O_3 (A, H and X) were assessed on the basis of the critical review by Konings et al. [22]. Interaction parameters were assessed for the liquid, DHCP α-(Nd,O) and BCC β-(Nd,O) to represent the oxygen solubility data in theses phases. A zoom of this area is given in Fig. 6b. Phase boundary data are well represented by the model. The model and the assessed parameters are reported in the Appendices A and B.

3.2.3. U–Nd–O ternary system

In a preliminary step [17], a tentative ternary phase diagram was calculated without including the experimental data (tielines in the miscibility gap for the FCC phase) determined in this work. The results evidenced that the existence of a miscibility gap in the FCC phase of the U–Nd–O system was possible on the basis of a thermodynamic model. This paper describes an improved thermodynamic modelling in which interaction parameters in the $(U_{1-y},Nd_y)O_2$ fluorite phase were optimised using the PARROT module on the basis of our experimental results and some data from the literature [15].

The available experimental data in literature on U–Nd–O system are given in Table 6. Concerning the thermodynamic data,

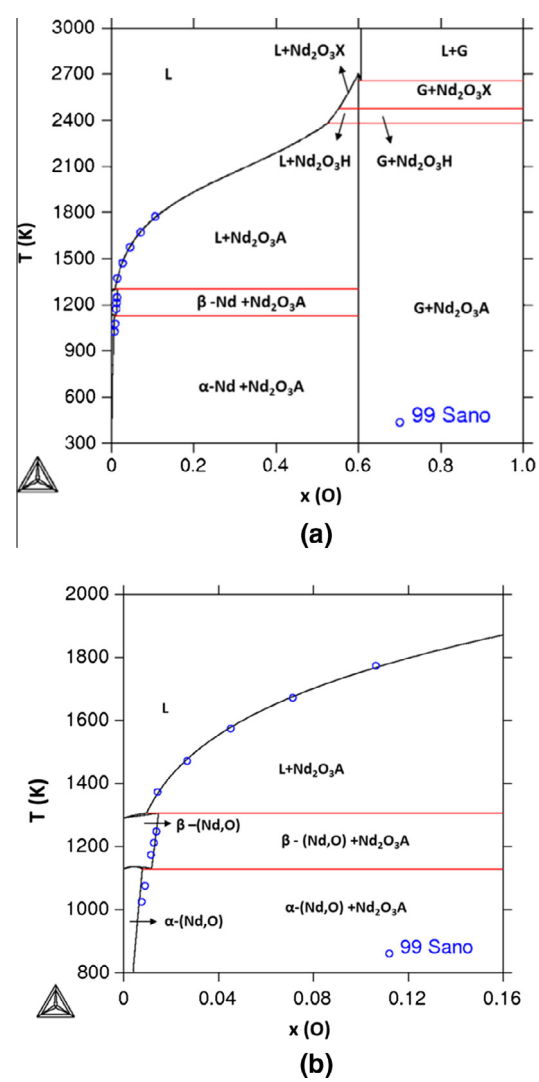

Fig. 6. Calculated Nd–O phase diagram: (a) for $0 < x(\text{Nd}) < 1$, (b) detail for $0 < x(\text{Nd}) < 0.16$.

Table 6
Available experimental data on $(U_{1-y}Nd_y)O_2$.

Available experimental data Heat capacity and enthalpy increments					
Authors	Temperature	y_{Nd}	Method	Used for assessment	References
Venkata	298–800 K	0.2–0.8	DSC Drop calorimetry	Yes	[3]
Oxygen potential					
Tetenbaum	2200–2300 K	0.1–0.2	Transpiration technique	No	[23]
Une	1123–1773 K	0.14–0.27	Thermogravimetry	Yes	[24]
Wadier	1123 K	0.1–0.5	Thermogravimetry E.m.f analyses	No	[25]
Phase diagram					
Our data (this paper)	Room temperature	0.01–0.28	XRD	Yes	This work
Our data XRD–HT	700 K	0.28	HT-XRD	Yes	[7]
Venkata	Room temperature	0.2–0.8	XRD	No	[3]
Wadier	1123 K	0–0.4	XRD	No	[25]
Boroujerdi	1523 K	0–0.9	XRD	No	[28]
Lambertson	1973–2773 K	0.2–1	XRD	Yes	[26]

we used recent experimental enthalpy increment data measured by Venkata [3] for $(U_{1-y},Nd_y)O_{2-x}$ solutions (from $y = 0.2$ to $y = 0.8$ Nd) using calorimetry. Oxygen potential measurements were measured by Tetenbaum [23], Une [24] and Wadier [25]. As the three dataset were found to be inconsistent with each other, only the oxygen potential data measured by Une were selected for the optimisation procedure in the 1123–1773 K temperature range and y_{Nd} values ranging from 0.14 to 0.27. In fact, the points measured by Tetenbaum [23] at high temperatures are too low compared to those measured by Une [24], which was already commented by Une, so these measurements were excluded as well as those of Wadier [25] for O/metal ratio below 2 with too low values

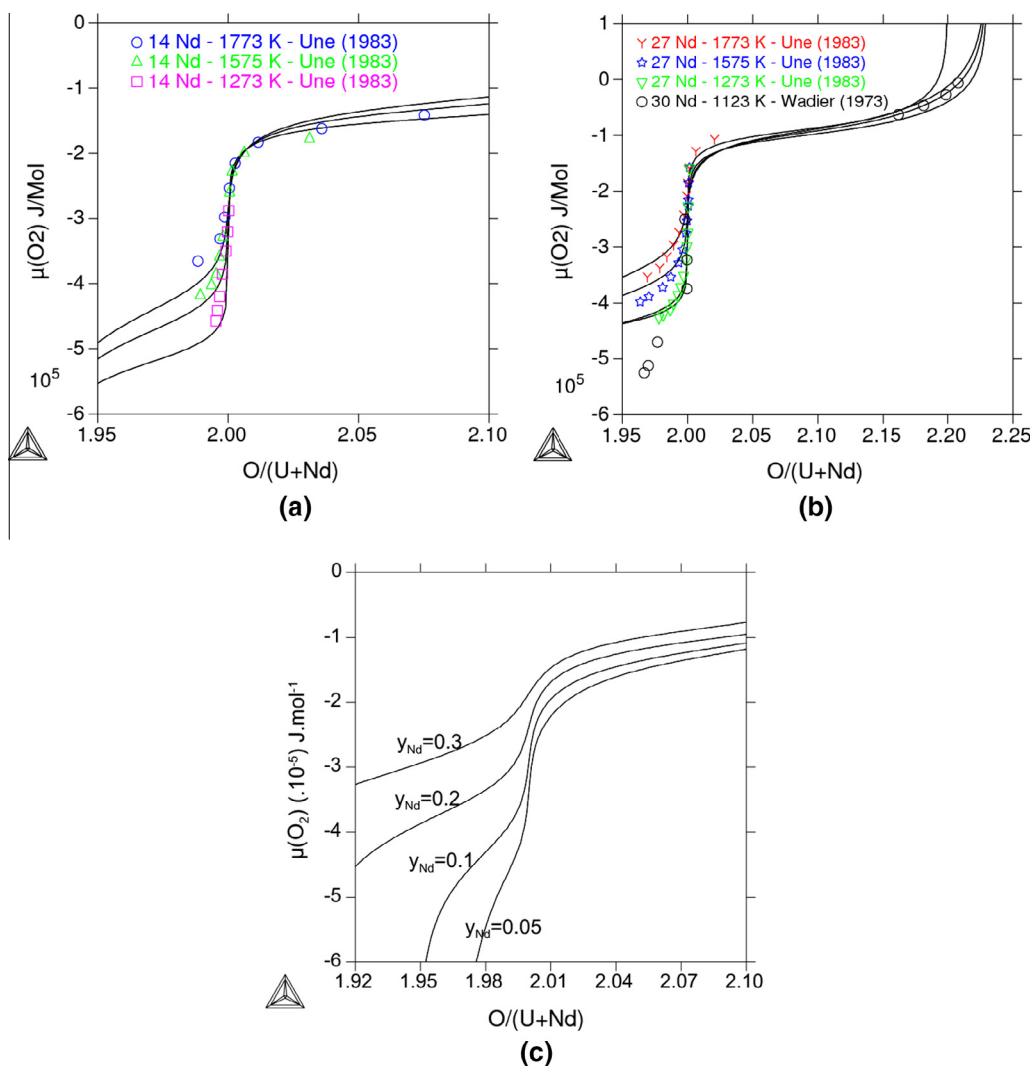

Fig. 7. Calculated oxygen potential versus O/(U + Nd) ratio, comparison with the data of Une [24] and Wadier [25]: (a) in $(U_{0.86}Nd_{0.14})O_2$ at 1273, 1575, 1773 K, (b) in $(U_{0.73}Nd_{0.27})O_2$ at 1273, 1575, 1773 K and in $(U_{0.7}Nd_{0.3})O_2$ at 1123 K, (c) for different y_{Nd} values at the sintering temperature of 1973 K.

which were not possible to reproduce with our model. This could be attributed to the fact that new Nd-enriched phases may form in this Nd composition range.

The phase diagram data used were mainly the experimental tielines in the FCC miscibility gap determined in this work, the transition temperature of the miscibility gap (FCC = FCC 1 + FCC 2) measured by us using HT-XRD already presented in [7], and the tentative phase diagram UO_2-Nd_2O_3 proposed by Lambertson and Mueller [26] to fix the boundary between FCC and (FCC + Nd_2O_3) region and liquidus data. Keller and Boroujerdi [27,28] investigated the UO_2–Nd_2O_3–UO_3 phase diagram at 1523 K in which several phase regions were found: single phase $(U_{1-y},Nd_y)O_{2+/-x}$ (FCC), two phase FCC + U_3O_8, two phase FCC + Rhombohedral (UO_2-6$NdO_{1.5}$–UO_3-6$NdO_{1.5}$), two phase Rhombohedral + Nd_2O_3-A. In our thermodynamic description, the Rhombohedral phase is not modelled.

The results of the oxygen potential calculations are shown in Fig. 7a and b. The isothermal section of the phase diagram calculated at 300 K and 800 K are shown in Fig. 8a and b for $0 < x_{Nd} < 0.4$ and $0.6 < x_O < 0.75$, and the isopleth section calculated for U/Nd = 2.595 is presented in Fig. 8c. Experimental points obtained in this work are represented, as well as the phase transition temperature determined in [7]. The thermodynamic models and the assessed parameters for the phases are reported in respectively Appendices A and B.

3.2.4. Comparison with literature

The calculated oxygen potential data are calculated for respectively Nd/(U + Nd) = 0.14 at 1273, 1575, 1773 K in Fig. 7a and for Nd/(U + Nd) = 0.27 at 1273, 1575, 1773 K and for Nd/(U + Nd) = 0.3 at 1123 K in Fig. 7b in comparison with Une's experimental data (for Nd/(U + Nd) = 0.14 and 0.27) and Wadier's data (for Nd/(U + Nd) = 0.3).

For O/metal ratio below 2, a satisfactory agreement was obtained with Une's experimental data for Nd content with y_{Nd} ranging from 0.14 to 0.3. And a good agreement was found with Wadier's data for O/metal ratio above 2. For higher Nd contents with y_{Nd} of 0.4 and 0.5, it was not possible to achieve a good agreement with Wadier's data for O/metal ratio below 2, due to the fact that these data are not consistent with Une's data. As explained previously, this could be attributed to the fact that new Nd-enriched phases may form in this Nd composition range, which are not considered in our calculations due to the absence of relevant experimental data. Nevertheless, a reasonnable agreement was found with Wadier for O/metal ratio higher than 2. Furthermore, Wadier reported that the FCC is very unstable at O/M ratios

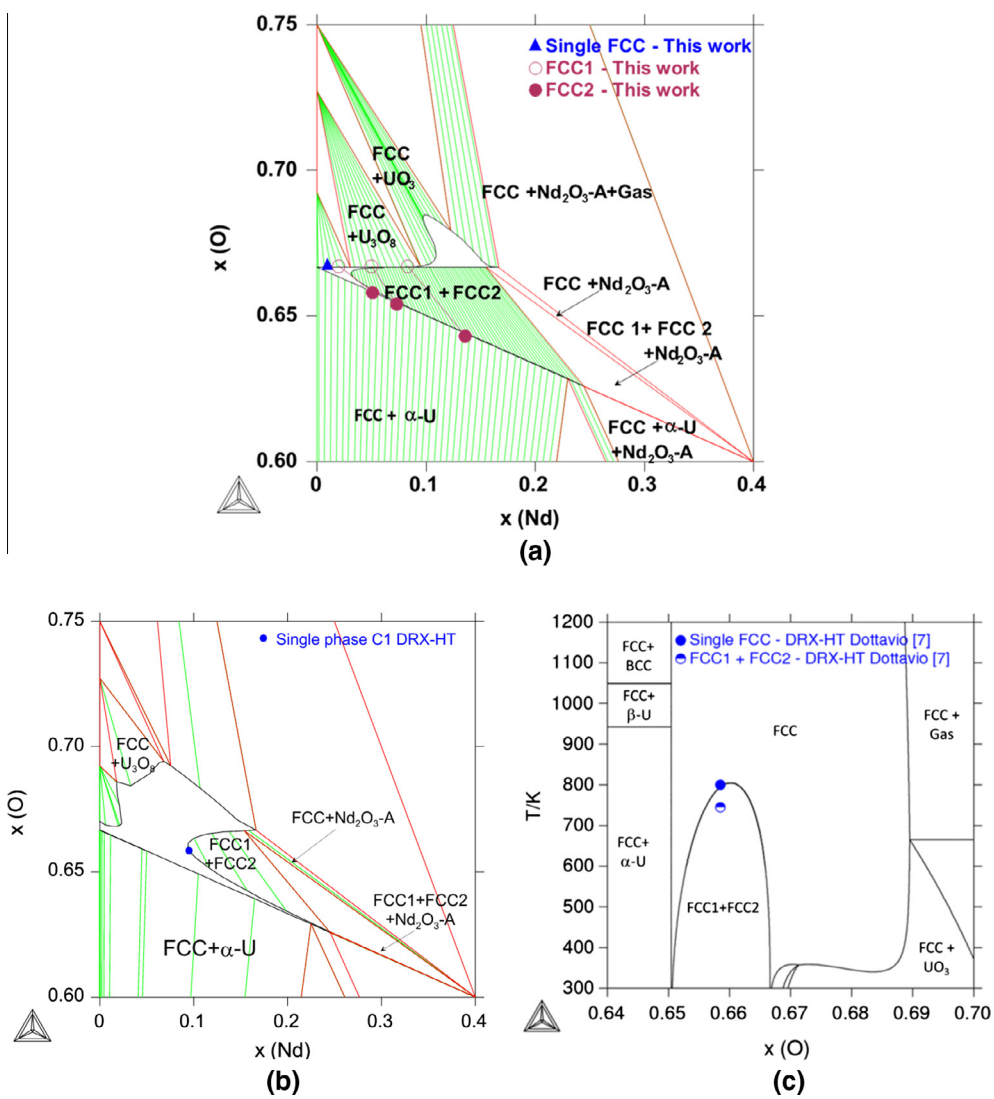

Fig. 8. Calculated U–Nd–O phase diagram (a) for $0 < x(Nd) < 0.4$ and $0.6 < x(O) < 0.75$ at 300 K (b) for $0 < x(Nd) < 0.4$ and $0.6 < x(O) < 0.75$ at 800 K (c) for U/Nd = 2.595. The experimental points obtained in the present work and in [7] are reported.

higher than two, when $y_{Nd} = 0.5$ which is well reproduced by our model. Nevertheless, experimental phase diagram data measured by Keller and Boroujerdi [27,28] show that the $(U_{1-y}Nd_y)O_2$ solution can extent up to 65% Nd at 1523 K. With our model, the maximum Nd content is 50%. The introduction of U^{6+} would be necessary to extent the composition range of the solid solution for higher Nd content. It has already suggested by McMurray [29], who reported similar problems for oxygen potential calculations using the Calphad method within the $(U_{1-y},Gd_y)O_2$ FCC phase for high Gd contents.

The theoretical assessment of the U–Nd–O system made it possible to evaluate the extent of the miscibility gap in the FCC phase as function of Nd and oxygen content as well as temperature, and to compare these calculated phase diagram data with our experimental results. At 300 K, an overall good agreement is found between the calculated and experimental data obtained in the present work, Fig. 8a. Nevertheless, for the sample B, an O/M = 2 was assumed for the oxygen enriched phase, but according to the calculation results in this composition range (where the miscibility gap closes) this phase should have a stoichiometry below 2. This can explain the discrepancies between the experimental data and the theoretical model at this point. At higher temperatures, the method used to determine tie-lines cannot be applied to the results of [7] because we do not consider the effects of lattice thermal expansion. However we can use the temperature phase transition obtained in [7] in order to optimise the miscibility gap assessment a high temperature. The results of the Fig. 8b and c showed a good agreement concerning this critical temperature (we assumed that the global O/M of the sample at 800 K is equal to the one of the sample at 300 K).

Concerning the phase diagram, we did not take into consideration the experimental data from authors who reported a single solid solution for $(U_{1-y}Nd_y)O_2$ oxides at low temperature because this is in contradiction with our results. Nevertheless, a good agreement is obtained with Wadier's data for samples quenched at 1123 K at which a large oxygen composition range is found where the mixed oxide is stable, even for large oxygen/metal ratios.

4. Discussion

This dual approach allowed us to obtain a consistent description of the miscibility gap existing in $(U_{1-y},Nd_y)O_2$ ceramics. The calculated phase diagram is in good agreement with our experimental measurements, for the tielines and the extent of the miscibility gap. However, in literature the existence e of a miscibility gap in the U–Nd–O system was not reported so far. This can be explained by two main reasons that we will discuss in the following.

First, in literature, the observation of two FCC phases is usually attributed to heterogeneity in the sample owing to a fabrication problem, and the assumption of a miscibility gap is generally disregarded. In order to analyse that point, we have studied a sample $(U_{0.72}Nd_{0.28})O_2$, issue from an incomplete sintering (sintering at 1673 K instead of 1993 K and pressed at 150 MPa). We performed an annealing treatment on this ceramic and we compared the results obtained with those reported in [7] for a sample of the same composition but with complete sintered. The annealing treatment was the same in both cases: the sample was maintained at 1973 K for 140 min, cooled down to room temperature at a very slow rate, and finally heated again to 1973 K.

In the sample analysed in [7], the XRD experiments in temperature have shown that the two FCC phases disappeared at 745 K during heating to produce a single solid solution, before system became biphasic again during cooling, and that in a reversible way, proving the existence of the miscibility gap. In the sample sintered at 1993 K, the evolution observed was not the same as shown on Fig. 9. Firstly because the system did not became a single solid solution at high temperatures, and secondly because the evolution in temperature was not reversible. Thus, we confirmed that this biphasic domain in this sample did not correspond to a miscibility gap but indeed to a fabrication problem.

The existence of a biphasic domain in the samples of [7] in truly due to the miscibility gap in U–Nd–O system because the results of [7] were not affected by the sample heterogeneity contrarily to the sample with incomplete sintering.

The second point discusses the fact that some authors did not detect this miscibility gap although they performed a structural

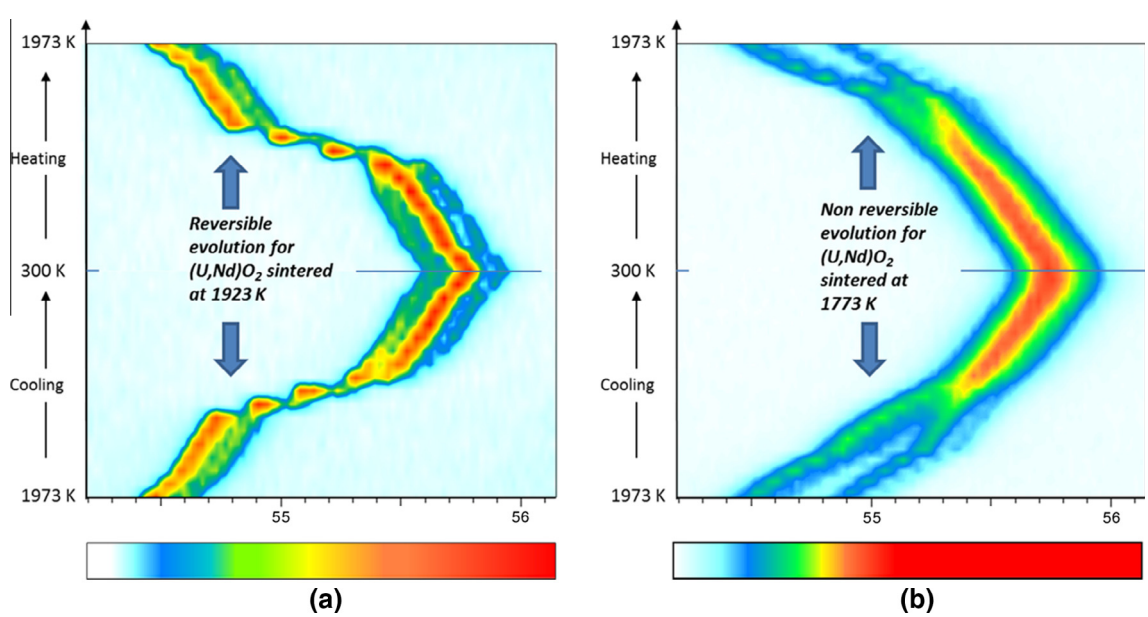

Fig. 9. Isodensity maps of XRD during the temperature variations. (a) $(U_{0.72}Nd_{0.28})O_2$ sintered at 1973 K, (b) $(U_{0.72}Nd_{0.28})O_2$ sintered at 1773 K. Evolution in (a) is reversible in temperature, but it not in (b). (Discontinuities observed in (a) at about 1100 K are due to the fact that the XRD measurement intervals were larger in this range).

analysis of the $(U_{1-y}Nd_y)O_2$ system, e.g. the recent work of Venkata [3]. As determined in this work, the miscibility gap appears when the overall O/M ratio of the system is less than two. The oxygen potential of the system during sintering is the main factor determining the final O/M content of the fuel (considering fast cooling after sintering). In Fig. 7c the oxygen potential as a function of O/M ratio at the typical sintering temperature of 1973 K are showed for different Nd contents. Based on our calculated oxygen potential data, it can be demonstrated that the oxygen partial pressure pO_2 of the system during sintering can be a critical factor influencing the appearance of the gap. It can be observed, as an example, that for an y_{Nd} of 0.2, the O/M ratio will be lower than two only if the oxygen potential during fabrication is lower than −300 kJ/mol. Furthermore, at higher Nd contents, the oxygen potential remains flat (it does not vary much with O/M ratio), which makes the control of the final O/M difficult. Consequently depending on the system's oxygen potential during fabrication this miscibility gap can or not be observed.

5. Conclusion

The miscibility gap in $(U_{1-y}Nd_y)O_2$ has been properly characterised and analysed. The results obtained from the thermodynamic modelling are consistent with the experimental observations. The gap appears preferentially when the O/M ratio is less than two and when the x(Nd) is higher than 0.04. Moreover, a method was proposed to distinguish whether a biphasic domain in $(U_{1-y}Nd_y)O_2$ results from a miscibility gap or an incomplete fabrication. It was also demonstrated that the stability of this gap is very sensitive to the pO_2 conditions of this system during fabrication. Some inconsistent results remain at high Nd concentrations, which will thus require more experimental data to improve the thermodynamic modelling of the U–Nd–O system. Consequently, oxygen potentials measurements during the gap appearance are expected to be conducted in a future experiment.

Acknowledgements

This work has been funded by the PERCOX project at the CEA in collaboration with Electricité de France and AREVA NP. The authors would like to thank J- Christophe Dumas, Hélène Rouquette, Catherine Tanguy, Marina Pontillon, Guillaume Volle, Vincent Marty and Jean-Christophe Richaud for their collaboration in this work.

Appendix A

The thermodynamic models are briefly presented in the following table. The thermodynamic parameters for the phases are reported in Appendix B.

Thermodynamic model for the Gibbs energy functions

(a) Pure elements and stoichiometric compounds

For each element i, the molar Gibbs energy is expressed by:

$$G_i^{\phi}(T) - H_i^{SER} = a + bT + cT\ln(T) + \sum_n d_n T^n$$

with $n = 2, 3, -1$

The Gibbs energy functions are referred to the enthalpy of the stable state of i at 298.15 K and 10^5 Pa (H_i^{SER}).

For the stoichiometric compounds, the molar Gibbs energy for one mole of formula unit, i.e. $\sum_i b_i^{\phi}$ moles of atoms is expressed by:

$$G_m^{\phi} - \sum_i b_i^{\phi} H_i^{SER} = a_0 + a_1\,T + a_2\,T\ln(T) + a_3\,T^2 + a_4\,T^{-1} + a_5\,T^3$$

where b_i^{ϕ} is the stoichiometric factor of element i in φ.

(b) Liquid phase

For the liquid, a two sublattice ionic liquid model [30,31] is used, with the following constituents:

$$(U^{4+}, Nd^{3+})_P(O^{2-}, Va^{Q-}, O)_Q$$

To maintain electroneutrality, charged vacancies are introduced in the second sublattice.

P and Q are the average charges on the first and second sublattices respectively:

$$P = 2y_{O^{2-}} + Qy_{VA^{Q-}}$$

$$Q = 4y_{U^{4+}} + 3y_{Nd^{3+}}$$

where y_i are the site fractions of the species i on the different sublattices.

The corresponding Gibbs energy function is:

$$G^{liq} - \sum_i x_i H_i^{SER} = {}^{ref}G^{liq} + {}^{id}G^{liq} + {}^{ex}G^{liq}$$

With $${}^{ref}G^{liq} = \sum_C\sum_A y_C y_A^{\circ} G^{liq}_{(C)(A)} + Q\ y_{Va}\sum_C y_C^{\circ} G_C^{liq} + Q\sum_B y_B^{\circ} G_B^{liq}$$

$${}^{id}G^{liq} = RT\left[P\sum_C y_C \ln y_C + Q\left(\sum_A y_A \ln y_A + y_{Va}\ln y_{Va} + \sum_B y_B \ln y_B\right)\right]$$

$$\begin{aligned}{}^{ex}G^{liq} = {} & \sum_{C_1}\sum_{C_2}\sum_A y_{C_1}y_{C_2}y_A L^{liq}_{(C_1,C_2)(A)} + \sum_{C_1}\sum_{C_2} y_{C_1}y_{C_2}y_{Va}^2 L^{liq}_{(C_1,C_2)(Va)} \\ & + \sum_C\sum_{A_1}\sum_{A_2} y_C y_{A_1} y_{A_2} L^{liq}_{(C)(A_1,A_2)} + \sum_C\sum_A y_C y_A y_{Va} L^{liq}_{(C)(A,Va)} \\ & + \sum_C\sum_A\sum_B y_C y_A y_B L^{liq}_{(C)(A,B)} + \sum_C\sum_B y_C y_B y_{Va} L^{liq}_{(C)(Va,B)} \\ & + \sum_{B_1}\sum_{B_2} y_{B_1} y_{B2} L^{liq}_{(B_1,B_2)}\end{aligned}$$

where A are anions, C cations and B neutral species.

(c) BCC phase: β-(Nd,O), γ-U

The BCC solid solution in U–Nd and Nd–O systems is described using the following two-sublattice model $(U,Nd)_1(O,Va)_3$ and the Gibbs energy of one mole of formula unit is expressed by:

$$\begin{aligned}G_m^{BCC} - \sum_i x_i H_i^{SER} = {} & \sum_i y_i y_{Va}^{\circ} G^{BCC}_{(i)(Va)} + \sum_i y_i y_O^{\circ} G^{BCC}_{(i)(O)} + RT\sum_i y_i \ln y_i \\ & + 3\,RT\sum_j y_j \ln y_j + \sum_i y_{i1} y_{i2} y_{Va} L^{BCC}_{(i1,i2)(Va)} \\ & + \sum_i y_i y_O y_{Va} L^{BCC}_{(i)(O,Va)}\end{aligned}$$

where i = U or Nd in the first sublattice and j = O or Va in the second sublattice. The second sublattice corresponds to the interstitial site in the BCC phase, where O is introduced as an interstitial specie and "Va" denote vacancy.

(d) DHCP phase: α-(Nd,O)

This phase is modelled to take into account the solubility of oxygen in the DHCP phase (α-Nd). The sublattice model is: $(Nd)_1(O,Va)_{0.5}$ and the Gibbs energy expression is given by:

$$\begin{aligned}G_m^{DHCP} - \sum_i x_i H_i^{SER} = {} & y_{Va}^{\circ} G^{DHCP}_{(Nd)(Va)} + y_O^{\circ} G^{DHCP}_{(Nd)(O)} + 0.5\,RT\sum_j y_j \ln y_j \\ & + \sum_i y_O y_{Va} L^{DHCP}_{(Nd)(O,Va)}\end{aligned}$$

The second sublattice corresponds to the interstitial site in the DHCP phase, where O is introduced as an interstitial specie and "Va" denote vacancy.

(e) C1 phase $(U_{1-y}Nd_y)O_2$

This mixed oxide is described by the three sublattice model using CEF [12]:

$$(U^{3+}, U^{4+}, U^{5+}, Nd^{3+})_1(O^{2-}, Va)_2(O^{2-}, Va)_1$$

The first sublattice is the site for cations with different possible oxidation states. The second one is the normal site for oxygen in the fluorine structure. Vacancies "Va" are introduced to describe hypo-stoichiometric compositions (O/(U + Nd) < 2). The third sublattice is the interstitial site for oxygen to represent hyper-stoichiometric compositions (O/(U + Nd)>2). The electroneutrality is maintained by changing the oxidation state of the cations in the first sublattice.

The Gibbs energy for this phase is described by:

$$G_m^{(U,Nd)O_2} = \sum_i \sum_j \sum_k y_i y_j y'_k \circ G^{(U,Nd)O_2}_{(i)(j)(k)} + RT\left(\sum_i y_i \ln y_i + 2\sum_j y_j \ln y_j + \sum_k y'_k \ln y'_k\right) + {}^{exc}G^{(U,Nd)O_2}$$

i = constituents of the first sublattice,
j = constituents of the second sublattice,
k = constituents of the third sublattice (as the second and third sublattices have the same species, the notation y' is used for the third one).

In the compound energy formalism, all the compounds formed by taking one species in each sublattice have to be described, even if they do not exist. This is the case for the fluorite oxide NdO_2 which is not a stable phase. The following Gibbs energy equation is proposed for this compound:

$${}^{\circ}G_{(Nd^{3+})(O^{-2})_2(Va)} = 0.5^{\circ}G_{Nd_2O_3} + 0.5^{\circ}G_{1/2}O_2 + 1.12467\,RT + C$$

$${}^{\circ}G_{(Nd^{3+})(O^{-2})_2(O^{-2})} = 0.5^{\circ}G_{Nd_2O_3} + 1.5^{\circ}G_{1/2O_2} + 1.12467\,RT + C$$

$${}^{\circ}G_{(Nd^{3+})(Va)_2(Va)} = 0.5^{\circ}G_{Nd_2O_3} - 1.5^{\circ}G_{1/2O_2} + 1.12467\,RT + C$$

$${}^{\circ}G_{(Nd^{3+})(Va)_2(O^{-2})} = 0.5^{\circ}G_{Nd_2O_3} - 0.5^{\circ}G_{1/2}O_2 + 1.12467\,RT + C$$

where "C" has a positive value.

Interaction parameters were assessed to fit the experimental miscibility gap and oxygen potential data:

$$\begin{aligned}{}^{exc}G^{(U,Nd)O_2} = {} & y_{Nd^{3+}} y_{U^{5+}} y_{O^{2-}} y'_{O^{2-}}\, {}^0L_{(Nd^{3+},U^{5+})(O^{2-})_2(O^{2-})} \\ & + y_{Nd^{3+}} y_{U^{5+}} y_{O^{2-}} y'_{Va}\, {}^0L_{(Nd^{3+},U^{5+})(O^{2-})_2(Va)} \\ & + y_{Nd^{3+}} y_{U^{4+}} y_{O^{2-}} y'_{Va}\, {}^0L_{(Nd^{3+},U^{4+})(O^{2-})_2(Va)} \\ & + y_{Nd^{3+}} y_{U^{4+}} y_{O^{2-}} y_{Va} y'_{Va}\, {}^0L_{(Nd^{3+},U^{4+})(O^{2-},Va)_2(,Va)} \\ & + y_{Nd^{3+}} y_{U^{5+}} y_{O^{2-}} y'_{O^{2-}} y'_{Va}\, {}^0L_{(Nd^{3+},U^{5+})(O^{2-})_2(O^{2-},Va)} \\ & + y_{Nd^{3+}} y_{U^{4+}} y_{O^{2-}} y'_{O^{2-}} y'_{Va}\, {}^0L_{(Nd^{3+},U^{4+})(O^{2-})_2(O^{2-},Va)}\end{aligned}$$

Appendix B

The thermodynamic parameters for the U–O system come from Guéneau et al. [12]. The thermodynamic parameters for the pure elements are taken from [32]. The thermodynamics parameters for Nd_2O_3 are taken from [22].

Assessed thermodynamic parameters (J/mol)	
Liquid $(Nd^{3+},U^{4+})_P(O^{2-},Va,O)_Q$	$G^{liq}_{(Nd^{3+})2(O^{2-})3} - 2H^{SER}_{Nd} - 3H^{SER}_{O} = -1,788,019.5 + 1271.0665\,T - 205\,T\ln T$
	$L^0_{(Nd^{3+})(O^{2-},Va)} = 120,767 - 38.9\,T$
	$L^0_{(Nd^{3+},U^{4+})(Va)} = 60,000$
	$L^1_{(Nd^{3+},U^{4+})(Va)} = -10,000$
	$L^0_{(Nd^{3+},U^{4+})(O^{2-})} = -500,000$
	$L^1_{(Nd^{3+},U^{4+})(O^{2-})} = 150,000$
BCC_A2: β-(Nd,O), γ-U	$G^{BCC}_{(Nd)(O)3} - H^{SER}_{Nd} - 3H^{SER}_{O} =^{\circ} G^{BCC}_{Nd} + 3^{\circ}G^{SER}_{O}$
$(Nd,U)_1(O,Va)_3$	$L^0_{(Nd,U)(Va)} = 70,000$
	$L^0_{(Nd)(O,Va)} = -1,751,674.40 + 360.7\,T$
DHCP: α-(Nd,O)	$G^{DHCP}_{(Nd)(O)0.5} - H^{SER}_{Nd} - 0.5H^{SER}_{O} =^{\circ} G^{SER}_{Nd} + 0.5^{\circ}G^{SER}_{O}$
$(Nd)_1(O,Va)_{0.5}$	$L^0_{(Nd)(O,Va)} = -287,362.7 - 45.2\,T$
Nd_2O_3 A,H,X	$G^{Nd_2O_3_A} - 2H^{SER}_{Nd} - 3H^{SER}_{O} =^{\circ} G(\mathrm{Nd2O3_A}) = -1,847,287.16 + 641.10881\,T - 117.107\,T\ln T - 0.1406827\,T^2 + 62,922\,T^{-1}$
$(Nd^{3+})_2(O^{2-})_3$	$G^{Nd_2O_3_H} - 2H^{SER}_{Nd} - 3H^{SER}_{O} = -1,853,698.95 + 1177.14163\,T - 190\,T\ln T$
	$G^{Nd_2O_3_X} - 2H^{SER}_{Nd} - 3H^{SER}_{O} = -1,843,698.95 + 1173.10449\,T - 190\,T\ln T$
$(U_{1-y}Nd_y)O_{2\pm x}$ (C1)	$G^{C1}_{(Nd^{3+})(O^{2-})2(O^{2-})} - H^{SER}_{Nd} - 3H^{SER}_{O} = +.5^{\circ}G(\mathrm{Nd2O3_A}) + 1.5^{\circ}G^{SER}_{O} + 20,000 + 1.12467\,RT$
$(U^{3+},U^{4+},U^{5+},Nd^{3+})(O^{2-},Va)_2(O^{2-},Va)$	$G^{C1}_{(Nd^{3+})(Va)2(O^{2-})} - H^{SER}_{Nd} - H^{SER}_{O}$
	$= +.5^{\circ}G(\mathrm{Nd2O3_A}) - 0.5^{\circ}G^{SER}_{O} + 20,000 + 1.12467\,RT$
	$G^{C1}_{(Nd^{3+})(O^{2-})2(Va)} - H^{SER}_{Nd} - 2H^{SER}_{O}$
	$= +.5^{\circ}G(\mathrm{Nd2O3_A}) + 0.5^{\circ}G^{SER}_{O} + 20,000 + 1.12467\,RT$
	$G^{C1}_{(Nd^{3+})(Va)2(Va)} - H^{SER}_{Nd}$
	$= +.5^{\circ}G(\mathrm{Nd2O3_A}) - 1.5^{\circ}G^{SER}_{O} + 20,000 + 1.12467\,RT$
	$L^0_{(Nd^{3+},U^{5+})(O^{2-})2(O^{2-})} = -744,337 - 265.526\,T$
	$L^0_{(Nd^{3+},U^{5+})(O^{2-})2(Va)} = -391,852$
	$L^0_{(Nd^{3+},U^{4+})(O^{2-})2(Va)} = -500,000 + 43\,T$
	$L^0_{(Nd^{3+},U^{5+})(O^{2-})2(O^{2-},Va)} = 451,921$
	$L^0_{(Nd^{3+},U^{4+})(O^{2-},Va)2(Va)} = 137,803$
	$L^0_{(Nd^{3+},U^{4+})(O^{2-})2(O^{2-},Va)} = -479,906$

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